

Executive Summary - Maroon Cays Pitch

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Small Island Developing States such as The Bahamas remain heavily dependent on imported diesel, resulting in high electricity costs and infrastructure vulnerability. While national solar projects are expanding generation capacity, they do not address resilience testing, technology validation, or commercialization. At the same time, tourism-driven waste accumulation strains limited landfill capacity. Maroon Cays addresses energy insecurity and waste management through an applied island-based pilot model.

The Maroon Cays Sustainable Infrastructure Research Center will function as a live island-based pilot testbed integrating solar resilience testing, microgrid optimization, and waste-to-energy research within a single commercialization-oriented hub. Unlike mainland research laboratories, the center validates technologies under real tropical stressors including hurricanes, corrosion, and grid instability. This model reduces commercialization risk, accelerates industry validation cycles, and positions Maroon Cays as the first integrated solar-and-waste pilot center in the Caribbean focused on applied resilience.

Key risks include regulatory coordination, capital intensity, and climate exposure; these will be addressed through early public-private alignment, phased pilot development, and blended funding strategies. Finally, environmental stressors serve as a validation advantage rather than a liability.

The partnership framework for the Maroon Cays Sustainable Infrastructure Research Center is structured as a multi-sectoral collaboration hub that bridges the gap between laboratory research and real-world application. Academic anchors such as the University of Chicago, Stanford University, and the University of The Bahamas will establish research legitimacy, workforce development pipelines, and regional alignment in the Bahamas. Industry collaboration will prioritize resilient solar and microgrid innovators, including Schneider Electric, BoxPower, ESS Inc., Sunfolding, and Swift Solar, whose technologies require validation under hurricane-prone, high-salinity conditions. Waste-to-energy partners such as CaribShare and Streamline Integrated Energy will support pilot-scale circular infrastructure deployment, particularly in anaerobic digestion and biodiesel conversion. Partnerships will be sequenced across three priority verticals: academic anchors, modular solar and microgrid innovators, and waste-to-energy commercialization partners. They will have the initial focus on academic and solar pilots to establish technical credibility before scaling into integrated waste infrastructure.

By prioritizing solar resilience pilots, integrating waste-to-energy research, and sequencing partnerships strategically, the center can establish early technical credibility while building a scalable commercialization pathway across Small Island Developing States. We recommend phased implementation beginning with a microgrid-scale solar pilot and academic anchor partnership, followed by expansion into integrated waste-to-energy systems. With the right partnership sequencing and disciplined execution, Maroon Cays can define the next generation of applied island infrastructure innovation.

Citations

Inter-American Development Bank. (n.d.). *How the Bahamas is making strides to improve its electricity service.*

<https://www.iadb.org/en/blog/energy/how-bahamas-making-strides-improve-its-electricity-service>

Rocky Mountain Institute. (n.d.). *Islands energy program.*

<https://rmi.org/our-work/islands-energy-program/>

United Nations Office for Disaster Risk Reduction. (2022). *Global assessment report on disaster risk reduction 2022.*

<https://www.undrr.org/publication/global-assessment-report-disaster-risk-reduction-2022>

EyeWitness News. (n.d.). *Government signs power purchase agreement for 20MW solar plant at Blue Hills power station.*

<https://ewnews.com/government-signs-power-purchase-agreement-for-20mw-solar-plant-at-blue-hills-power-station/>

Eco Energy Bahamas. (n.d.). *Solutions.* <https://www.ecoenergybahamas.com/solutions>

National Renewable Energy Laboratory. (n.d.). *About NREL.* <https://www.nrel.gov/about/>

National Institute of Standards and Technology. (n.d.). *Performance of photovoltaic systems under extreme wind conditions.*

<https://www.nist.gov/publications/performance-photovoltaic-systems-under-extreme-wind-conditions>

de Oliveira, D. E. P., Miranda, A. C., Junior, M. V., Santana, J. C. C., Tambourgi, E. B., Facchini, F., Iavagnilio, R., & Pinto, L. F. R. (2024). Economic and environmental feasibility of cogeneration from food waste: A case study in São Paulo City. *Sustainability*, 16(7), 2979. <https://doi.org/10.3390/su16072979>

Sealey, K. S., & Smith, J. (2014). Recycling for small island tourism developments: Food waste composting at Sandals Emerald Bay, Exuma, Bahamas. *Resources, Conservation and Recycling*, 92, 25–37. <https://doi.org/10.1016/j.resconrec.2014.08.008>

U.S. Department of Energy. (n.d.). *Microgrid portfolio.*

<https://www.energy.gov/oe/activities/technology-development/grid-modernization-and-smart-grid/microgrid>

International Renewable Energy Agency.(2017). *Electricity storage and renewables: Costs and markets*.<https://www.irena.org/publications/2017/Oct/Electricity-storage-and-renewables-costs-and-markets>

Green Climate Fund. (n.d.). *Small island developing states portfolio*.
<https://www.greenclimate.fund/themes/small-island-developing-states>